successful model that unified the electromagnetic and weak interactions in the Standard Model of particle physics, proposed a clever modification of the Schrödinger equation that manages to maintain the total probability over time. It comes at a cost, however; the simplest version of Weinberg's theory allows you to send signals faster than light between entangled particles, as opposed to the no-signaling theorem of ordinary quantum mechanics. This flaw can be patched, but then something even weirder occurs: not only are there still other branches of the wave function, but you can actually send signals between them, building what physicist Joe Polchinski dubbed an "Everett phone." Maybe that's a good thing, if you want to base your life choices on the outcome of a quantum measurement and then check in with your alternate selves to see which one turned out the best. But it doesn't seem to be the way that nature actually works. And it doesn't succeed in solving the measurement problem or getting rid of other worlds.

In retrospect this makes sense. Consider an electron in a pure spin-up state. That can equally well be expressed as an equal superposition of spin-left and spin-right, so that an observation along a horizontal magnetic field has a 50 percent chance of observing either outcome. But precisely because of that equality between the two options, it's hard to imagine how a deterministic equation could predict that we would see either one or the other (at least without the addition of new variables carrying additional information). Something would have to break the balance between spin-left and spin-right.

We therefore have to think a bit more dramatically. Rather than taking the Schrödinger equation and gently tinkering with it, we can bite the bullet and introduce a completely separate way for wave functions to evolve, one that squelches the appearance of multiple branches. Plenty of experimental evidence assures us that wave functions *usually* obey the Schrödinger equation, at least when we're not observing them. But maybe, rarely but crucially, they do something very different.

What might that different thing be? We seek to avoid the existential horror of multiple copies of the macroscopic world being described in a single wave function. So what if we imagined that wave functions undergo occasional *spontaneous collapse*, converting suddenly from being spread out over different possibilities (say, positions in space) to being relatively well localized around just one point? This is the key new feature of